

Traveling the world

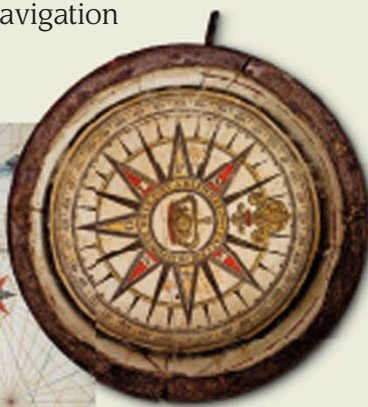
The word “navigation” originally meant finding your way at sea. For centuries, sailors would keep close to the coast, using landmarks and local knowledge of currents and weather conditions to navigate by. Later, they used compasses to indicate the direction in which to sail and they developed navigational aids to calculate their position at sea. Now we use the term navigation to mean finding your way anywhere.



Portolan chart of the Gulf of Mexico, 1547

Early navigational chart

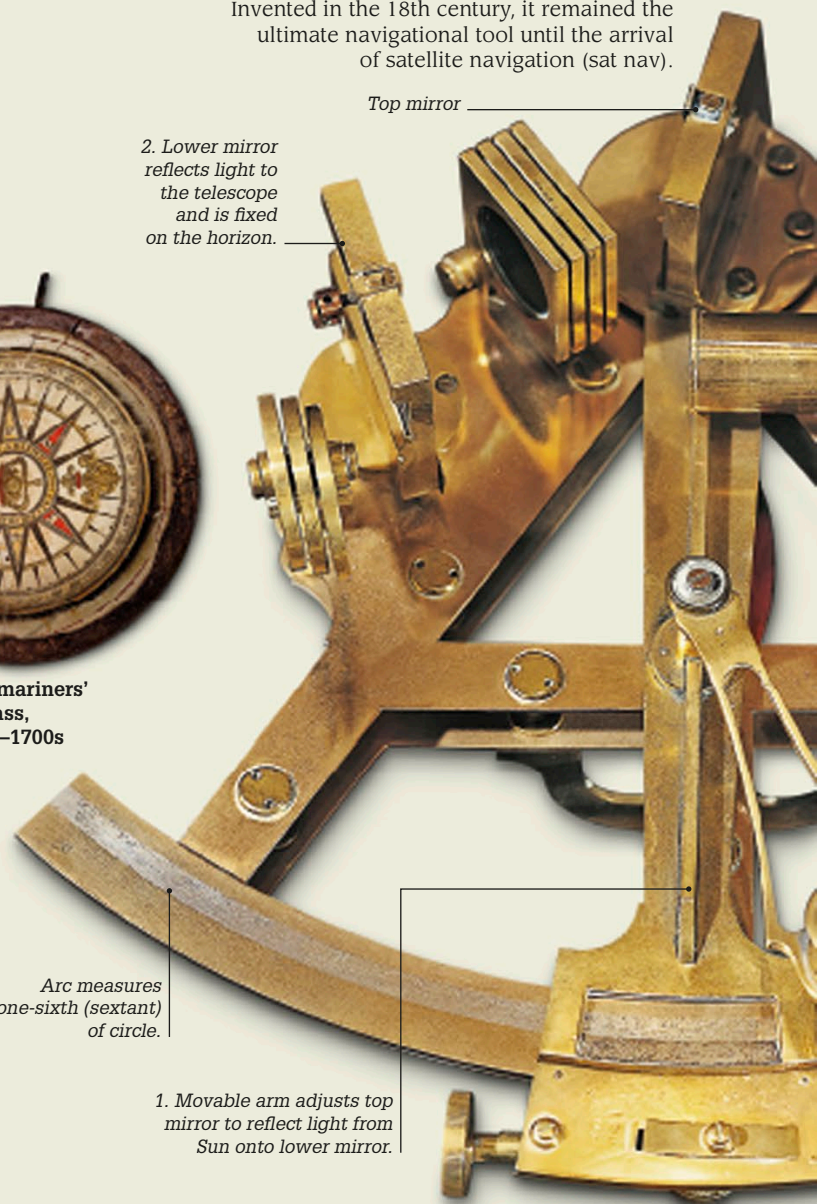
This portolan is drawn “upside-down” with south at the top. Places along the coast are carefully named and located, while inland areas contain fanciful scenes. Compass “roses” give direction. The captain plotted a course by following the lines that crisscross the chart from the roses.



Early mariners' compass, c 1500–1700s

Celestial navigation

The sextant was a highly accurate instrument used to determine latitude (position north or south of the equator) by measuring the angle between the horizon and the Sun during the day, or the Moon, planets, and stars at night. Invented in the 18th century, it remained the ultimate navigational tool until the arrival of satellite navigation (sat nav).



Top mirror

2. Lower mirror reflects light to the telescope and is fixed on the horizon.

Arc measures one-sixth (sextant) of circle.

1. Movable arm adjusts top mirror to reflect light from Sun onto lower mirror.

Key events

1000

The Vikings used a device called a sun compass (a wooden disc with directional markings) to help them navigate by the Sun.



Viking sun compass

1300

Knowledge of the magnetic compass reached Europe from China, where it had been in use for at least 1,000 years.

1400

European sailors began using coastal charts called portolans in combination with a compass to plot a course from port to port.

1569

Mercator's map projection, on which lines of latitude and longitude intersect at right angles, made it easier to navigate at sea.